

DPDK IDS Testing Workflow (New Pipeline)

Workflow Diagram



└ Metrics Dashboard (optional, port 5000) [Optional]

Command: `sudo bash run_realtime_engine_dpdk.sh start`

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STEP 3: Feed Traffic

Live Traffic via DPDK-bound NIC

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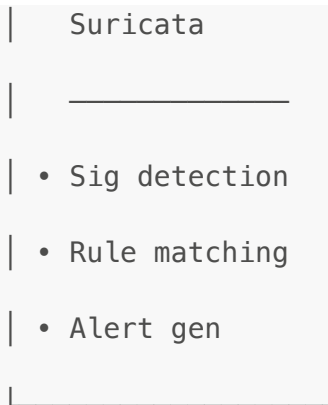
Optional PCAP Replay for testing

```
| replay_pcap_for_testing.py → Kafka (pcap-data)
|
| Example:
| python3 dpdk_suricata_ml_pipeline/src/replay_pcap_for_testing.py
|   --pcap-file dataset.pcap --ground-truth labels.csv --real-time
```

↓

STEP 4: IDS Processing (Automatic)

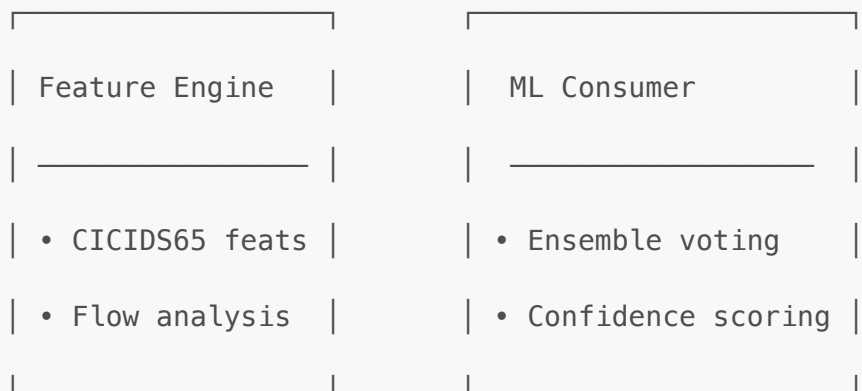
DPDK NIC RX → Suricata (DPDK) → Kafka



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Kafka Topic: suricata-alerts

DPDK NIC RX → Feature Engine → Kafka → ML Consumer → Kafka



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Kafka Topic: ml-features

↓

Kafka Topic: ml-

predictions

↓

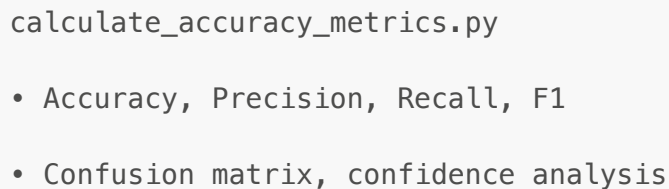
STEP 5: Accuracy Calculation (Optional)

Ground Truth (from PCAP)

Predictions (from ML Consumer)

labels.csv / packet metadata

logs/ml_predictions.csv



```
graph TD; A[ ] --> B["• Accuracy, Precision, Recall, F1<br/>• Confusion matrix, confidence analysis"]
```

calculate_accuracy_metrics.py

- Accuracy, Precision, Recall, F1
- Confusion matrix, confidence analysis

1. Configuration & Verification

- Ensure DPDK prerequisites and Suricata with DPDK support are installed.
- Bind the target NIC to DPDK and configure input.
- Update the central pipeline configuration.

Commands:

```
# Bind and configure NIC for DPDK input
bash dpdk_suricata_ml_pipeline/scripts/01_bind_interface.sh
bash dpdk_suricata_ml_pipeline/scripts/04_configure_dpdk_input.sh

# Edit pipeline settings (interfaces, Kafka, model paths)
vim dpdk_suricata_ml_pipeline/config/pipeline.conf
```

Key settings to review in `dpdk_suricata_ml_pipeline/config/pipeline.conf`:

- `NETWORK_INTERFACE` and `INTERFACE_PCI_ADDRESS`
- `KAFKA_BOOTSTRAP_SERVERS` and topic names
- `ML_MODEL_PATH` and `ML_FEATURES_CONFIG`

Example model path on macOS:

```
ML_MODEL_PATH="/Users/sujayschakravarthy/Programming/IDS/ML
Models/decision_tree_model_2017.joblib"
ML_FEATURES_CONFIG="/Users/sujayschakravarthy/Programming/IDS/config/ids
_config.yaml"
```

2. Start IDS Pipeline

- Start everything from the single runner.
- The runner orchestrates Kafka, Suricata (DPDK), Feature Engine, ML consumers, and optional dashboard.

Commands:

```
# Start full pipeline (requires sudo)
sudo bash run_realtime_engine_dpdk.sh start

# Check status
sudo bash run_realtime_engine_dpdk.sh status

# Stop all
sudo bash run_realtime_engine_dpdk.sh stop
```

Services started:

- Kafka ([localhost:9092](#))
- Suricata in DPDK mode (signature alerts → [suricata-alerts](#))
- Feature Engine (CICIDS features → [ml-features](#))
- ML Consumer (predictions → [ml-predictions](#))
- Suricata Alerts ML Consumer (optional → [ml-predictions](#))
- Metrics Dashboard (optional → [http://localhost:5000](#))

Code references:

- Runner start sequence: [run_realtime_engine_dpdk.sh:631-636](#)
- Feature Engine start: [run_realtime_engine_dpdk.sh:173-251](#)
- Ensemble ML consumer: [run_realtime_engine_dpdk.sh:253-342](#)
- Suricata alerts ML consumer: [run_realtime_engine_dpdk.sh:344-405](#)
- Dashboard start: [run_realtime_engine_dpdk.sh:407-438](#)

3. Feed Traffic

- Live capture: Send real traffic through the DPDK-bound NIC.
- Optional PCAP-based testing: Replay labeled PCAPs into Kafka for analysis.

PCAP replay (optional):

```
python3 dpdk_suricata_ml_pipeline/src/replay_pcap_for_testing.py \
  --pcap-file path/to/dataset.pcap \
  --ground-truth path/to/labels.csv \
  --real-time --speed 1.0
```

This publishes packet metadata to Kafka topic [pcap-data](#) for analysis and ground-truth correlation.

4. IDS Processing (Automatic)

- DPDK NIC → Suricata (DPDK) → Kafka `suricata-alerts`
- DPDK NIC → Feature Engine → Kafka `ml-features` → ML Consumer → Kafka `ml-predictions`

Relevant components:

- DPDK feature engine to `ml-features`:
`dpgk_suricata_ml_pipeline/src/dpgk_feature_engine.py`
- Ensemble/single model consumers:
`dpgk_suricata_ml_pipeline/src/realtime_ensemble_consumer_with_csv.py`,
`realtime_ml_consumer.py`
- Suricata alerts ML consumer: `dpgk_suricata_ml_pipeline/src/ml_kafka_consumer.py`

5. Accuracy Calculation (Optional)

- Use the CSV produced by the ensemble consumer for metrics.
- Produces JSON reports and optional confusion matrix PNG.

Commands:

```
python3 dpgk_suricata_ml_pipeline/src/calculate_accuracy_metrics.py \  
--predictions logs/ml_predictions.csv \  
--output logs/accuracy_report.json \  
--confusion-matrix logs/confusion_matrix.png
```

Quick Command Reference

- Configure NIC: `bash dpgk_suricata_ml_pipeline/scripts/04_configure_dpdk_input.sh`
- Start pipeline: `sudo bash run_realtime_engine_dpdk.sh start`
- Status: `sudo bash run_realtime_engine_dpdk.sh status`
- Stop: `sudo bash run_realtime_engine_dpdk.sh stop`
- Check Kafka topics: `kafka-topics.sh --list --bootstrap-server localhost:9092`
- View predictions: `kafka-console-consumer.sh --topic ml-predictions --bootstrap-server localhost:9092`

Monitoring

- Feature Engine log: `tail -f logs/feature_engine.log`
- ML Consumer log: `tail -f logs/ml_consumer.log`
- Suricata log: `tail -f /var/log/suricata/suricata.log`
- Dashboard: `http://localhost:5000`

Notes

- Always run the runner with `sudo` to access raw sockets and DPDK.

- Ensure `pipeline.conf` points to valid model files in `ML Models/` and the correct Kafka broker.
- For AF_PACKET testing instead of DPDK, use `dpdk_suricata_ml_pipeline/src/realtime_feature_engine.py` with `sudo`.